Mohammed Yusuf

System Engineer — Java Backend

+91 80566 60332 | mohammedyusuf1799@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Backend Software Engineer with 4+ years of experience designing and delivering high-availability, distributed Java and Spring Boot systems at enterprise scale. Proven track record in event-driven microservices architecture with Apache Kafka, test-driven development with SonarQube quality gates, and CI/CD pipeline automation. Hands-on with relational databases, observability tooling, and production infrastructure — and builds self-hosted agentic AI systems with LLMs, RAG, and tool-calling. Microsoft Azure Security Engineer certified.

Technical Skills

Languages	Java, Python, SQL
Backend	Spring Boot, Spring Framework, Spring Data, Spring Security, Hibernate / JPA, Apache Kafka, REST API, Microservices, Event-Driven Architecture
Testing & Quality	JUnit, Mockito, TDD, Integration Testing, SonarQube, Code Review
DevOps & CI/CD	Docker, Jenkins, GitHub Actions, Maven, Linux, Oracle Cloud (OCI)
Databases	PostgreSQL, MySQL, SQLite, NoSQL (familiarity)
Observability	Prometheus, Grafana, Alertmanager, Distributed Tracing, Structured Logging

Experience

Tata Consultancy Services — Bengaluru, India

System Engineer — Java Backend — Aug 2022 – Present

- Designed and developed Java and Spring Boot microservices for mission-critical financial transaction platforms, engineering 40+ REST API endpoints and improving application execution speed by **35%**.
- Architected an event-driven, high-throughput pipeline using Apache Kafka, replacing synchronous REST calls and reducing P99 processing latency by **35%** while improving system throughput by **40%** — handling peak financial transaction loads with high availability.
- Implemented TDD using JUnit and Mockito with high unit test coverage and SonarQube quality gates enforced on every release — reducing production defects by **30%** across consecutive releases.
- Built and maintained CI/CD pipelines with Docker and Jenkins, reducing build and deployment cycle times by **20%** and enabling continuous delivery across test, integration, and production environments.
- Led migration of legacy .NET and mainframe systems to a scalable, distributed Spring Boot microservices architecture, delivering zero data loss during cutover across multiple production services.
- Conducted regular code reviews, drove continuous refactoring, and mentored junior engineers on Java best practices, accelerating team delivery velocity by 20%.
- Designed and documented RESTful API specifications, system architecture diagrams, and technical design documents for cross-functional teams, ensuring alignment between business requirements and scalable software solutions.
- Collaborated with product and QA teams in Agile/Scrum sprints — participating in story refinement, sprint planning, and retrospectives — consistently delivering features on time with zero critical post-release incidents.

Projects

Falak AI — Agentic AI Execution System (self-built) — 2025 – Present

Python, Ollama, Anthropic API, ChromaDB, FastAPI

- Built a private, self-hosted agentic AI system from scratch: an agentic loop with planning, tool-calling, and ReAct-style execution over 8 tools (shell, file, web search, Notion, n8n).
- Engineered smart model routing — complexity scoring sends simple queries to a local LLM (Ollama) and complex ones to the Anthropic API — minimizing cost while keeping data private.
- Implemented a 3-tier memory system (SQLite sessions + ChromaDB semantic recall + typed memory files) with RAG ingestion, auto-memory extraction, and context compaction.
- Shipped a streaming Telegram interface (plus FastAPI REST + CLI) with prompt-injection detection, an audit log, allowlist access control, and human-in-the-loop approval for destructive actions.

Falak — Self-Hosted Infrastructure & Automation — 2025 – Present

Docker, Traefik v3, Prometheus, Grafana, Tailscale, n8n

- Designed and operate 20+ Dockerized services across a home server and an Oracle Cloud VPS, with Traefik v3 reverse proxy, automated TLS via Let's Encrypt, and full observability (Prometheus, Grafana, Alertmanager with custom alert rules).
- Hardened with CrowdSec threat detection, Authelia SSO + TOTP 2FA, a default-deny firewall, and a Tailscale zero-trust mesh; zero open ports via Cloudflare Tunnel.
- Built n8n automation for AI-driven health monitoring, self-healing backup recovery, and Telegram-based alerting; daily encrypted + weekly AES-256 offsite backups.

Education & Certifications

AVS Engineering College — B.E. Mechanical Engineering, CGPA 7.49 — Tamil Nadu, 2016–2020

Microsoft Certified: Azure Security Engineer Associate — 2023–2027

Microsoft | Credential ID: C2BED8FD1B1373F

Claude Certified Developer - Foundations — 2026–2027

Anthropic | Credential